Unit Gateway H2

SKU:U195





















Description

Unit Gateway H2 is a brand-new form-factor gateway unit built around the **ESP32-H2-MINI-1-N2** module. It features a 32-bit single-core MCU based on the **RISC-V** architecture and integrates **IEEE 802.15.4** wireless communication, supporting **Zigbee**, **Thread**, and Thread-based **Matter** protocols. A built-in hardware encryption engine meets the security requirements of high-level IoT applications. The product communicates with M5 series hosts via the **Grove (Port C)** interface over UART, enabling data exchange. **Unit Gateway H2** can flexibly act as a gateway (e.g., Thread Border Router) or an end device, cooperating with a host to build cross-ecosystem connectivity or running IoT applications independently. A standard download header and **Type-C** port are reserved for firmware flashing and debugging of the H2 chip. Typical application scenarios include smart home, environmental monitoring, sensor networks, and low-power wireless communication nodes.

Features

- Supports wireless communication protocols such as Zigbee, Thread, and Matter
- Integrated IEEE 802.15.4 technology
- UART serial communication
- Hardware encryption support
- Low power consumption
  - Arduino IDE
  - ESP-IDF

Includes

- 1 x Unit Gateway H2
- 1 x HY2.0-4P Grove Cable (20 cm)

Applications

- Smart home
- Industrial automation
- Healthcare
- Consumer electronics
- Smart agriculture

Specifications

Specification	Parameter
SoC	Built-in ESP32-H2 module, 32-bit single-core RISC-V processor, up to 96 MHz @IEEE 802.15.4
Flash	2 MB
Antenna	On-board PCB antenna
Supported Protocol	IEEE 802.15.4 (including Zigbee 3.0, Thread 1.3, Matter)
Power Consumption	Operating current: (Thread network) DC 5 V @ 26.08 mA Standby current: DC 5 V @ 16.16 mA Sleep current: (Grove supply) DC 5 V @ 92.7 µA
Operating Temp.	0 – 40 °C
Product Size	48.0 x 24.0 x 8.0 mm
Product Weight	6.9 g
Package Size	138.0 x 93.0 x 9.0 mm
Gross Weight	12.3 g

Learn

Download Mode

> Press and hold the Boot button, then insert the USB cable into the device. Release the Boot button after the connection is stable to enter download mode.

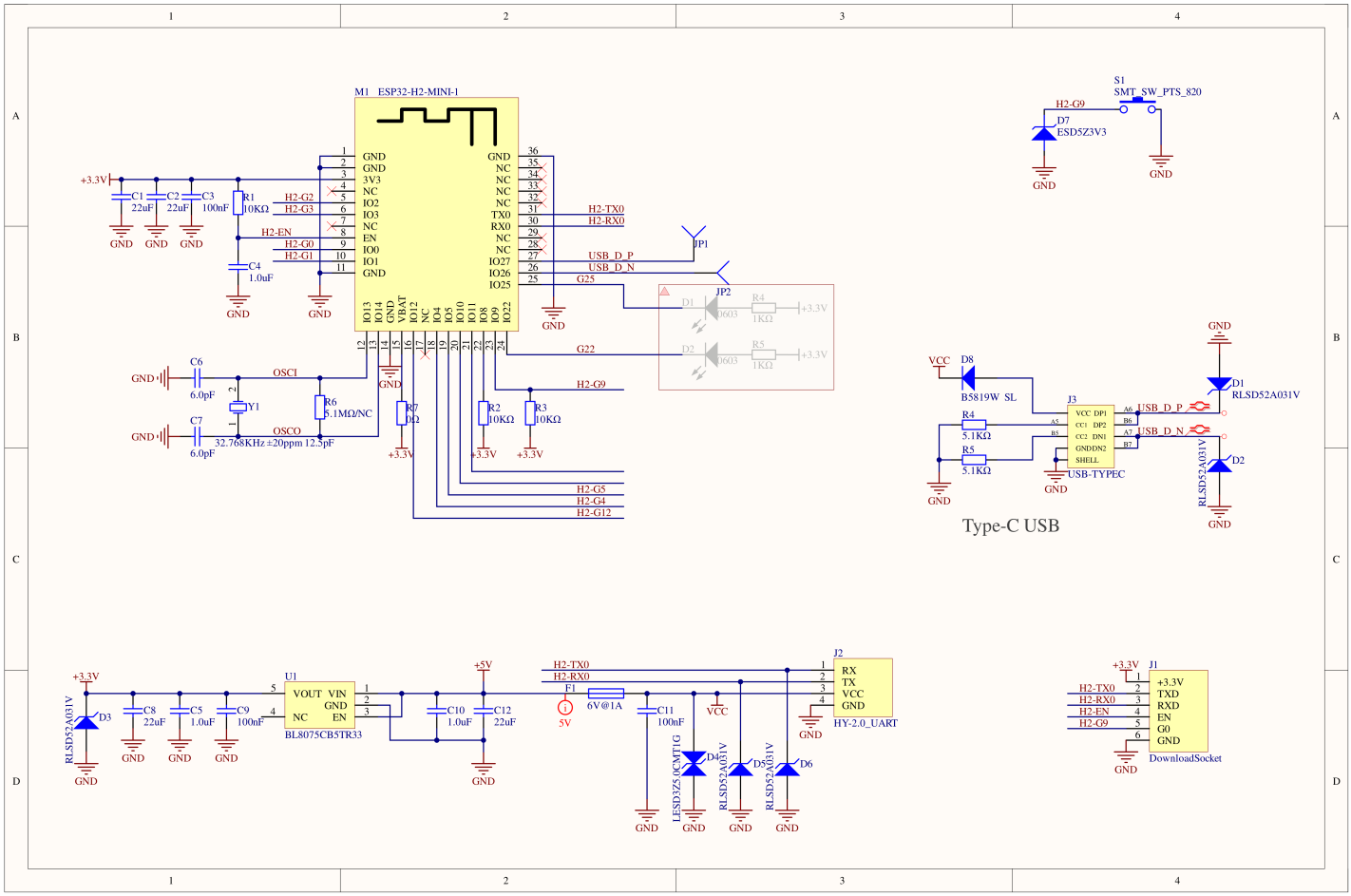


Certifications

- ESP32-H2-MINI-1 Module Certifications

Schematics

- Unit Gateway H2 Schematics PDF

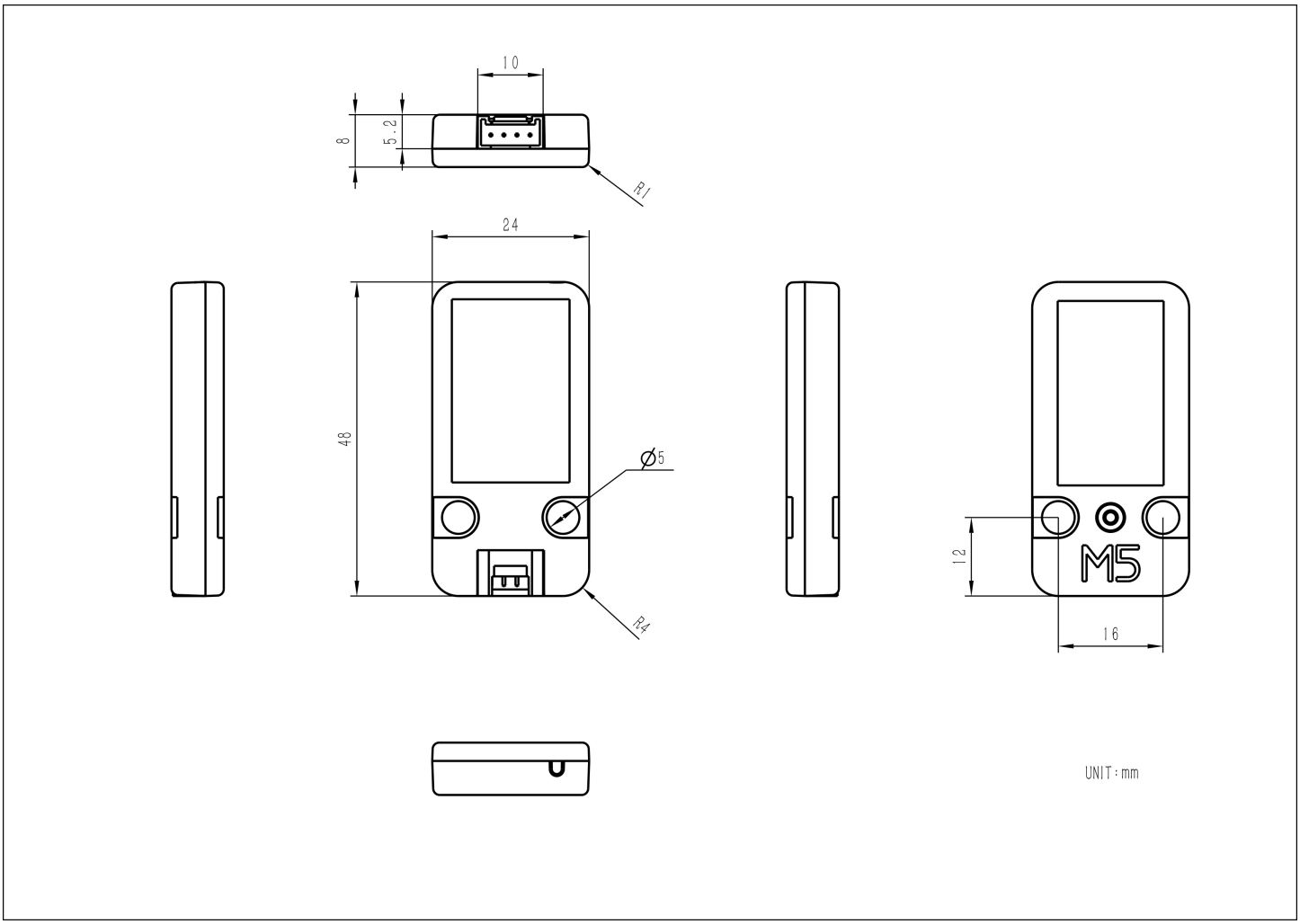


PinMap

Unit Gateway H2

HY2.0-4P	Black	Red	Yellow	White
PORT.C	GND	5 V	UART_RX	UART_TX

Model Size



Structure

- Unit Gateway H2 Structure Files

Datasheets

- ESP32-H2-MINI-1-N2 Module

Softwares

Arduino

- Unit Gateway H2 Arduino Guide

UiFlow2

- Unit GatewayH2 UiFlow2 Docs

ESP-IDF

- ESP Thread Border Router Guide
- ESP Zigbee Gateway Guide
- ESP Zigbee NCP Guide
- ESP32-H2 ESP-IDF Docs
- ESP Zigbee SDK

- ESP Thread Border Router SDK

Video

- Unit Gateway H2 product introduction and demonstration

U195_Unit_Gateway_H2_vodeo.mp4